

Lab 4: Basic Networking Commands in Linux OS and Windows OS

Lab Date: June 9th, 2026

Deadline: June 16th, 2026

Class: BSc.CSIT 4th Semester, New Summit College

Subject: Computer Network

Submission Mode: **Handwritten Document** (Print Output Section only)

Objective: Understand the basics of OS Installation & basic networking commands

Tasks:

- Understand the concepts of basic networking commands listed below.

Report Format & Structure (Topics to include for submission):

- Theory
 - a. **Networking Commands** (Explain the networking commands given below with the figure of working for each command)
 - i. For each command, include
 1. Description
 2. Use
 3. Syntax
 4. Figure
- Conclusion

Reference:

1. <https://drive.google.com/file/d/1iw30b3PgVY7CDEgzFfV2SISRih7i4a5/view?usp=sharing>
2. <https://drive.google.com/file/d/1RtryMWjRydXCgyePg7rFP657s3fWp7N0/view?usp=sharing>

List of Networking Commands: Linux vs Windows Syntax

Command	Linux Syntax	Windows Syntax	Purpose
ping	ping <hostname/IP>	ping <hostname/IP>	Test network connectivity
getmac	✗ (Not available)	getmac	Show MAC address of local interfaces
ipconfig	✗ (Use <i>ifconfig</i> or <i>ip</i>)	ipconfig / ipconfig /all	View IP configuration
hostname	hostname	hostname	Show system hostname
tracert	tracert <hostname>	tracert <hostname>	Trace packet route to destination
netstat	netstat -tulnp	netstat -an / netstat -b	View network connections and listening ports
nslookup	nslookup <domain>	nslookup <domain>	DNS lookup for a domain name
route print	route -n or ip route show	route print	View routing table
nmap	nmap <target>	nmap <target> (if installed)	Network scanner for hosts and ports
netcat	nc <target> <port>	nc <target> <port> (with ncat)	Raw TCP/UDP connections and port scanning
curl	curl <URL>	curl <URL> (Windows 10+)	Transfer data to/from a server

systeminfo	uname -a / lsb_release -a	systeminfo	System hardware and OS info
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